

Water and fluid dynamics in fascia

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Hydration keeps fascia alive

The human body consists mainly of water. In fact, the average adult person consists of up to 60% water. The amount of water, however, is not constant throughout the lifespan. When we are born, water constitutes approximately 90% of our body mass and as we age we gradually become less hydrated, until at over 80 years the water content may be as small as 50% (Watson et al., 1980). The main fluid of the body can be found in virtually any tissue: even in dense ligaments and tendons, the water content is >50%. We have observed this amazing fact many times in our laboratory when measuring the wet weight of ligaments from animals and then comparing this with the same weight after the tissue had been completely dried. The majority of our body water (~65%) is interstitial water, which is dissolved in our extracellular matrix. Most water enters the ground substance via the small arterioles, in which it was expressed as blood plasma. Once in the ground substance, the water usually binds in one way or another with the hydrophilic surfaces of the proteoglycans, until it may depart again, back into the central bloodstream via small lymphatic or venous vessels (Figure 9.1).

Note that only 10% of that return happens via the lymph vessels, while the remaining 90% occurs via the small venules (Schwartz et al., 2019). This may have clinical implications for fascia-oriented treatments like foam rolling, massage or vacuum treatment via suction cups.

While in healthy tissue a multidirectional motion may be the most efficient renewal approach for the ground substance water, in inflammatory conditions—such as in oedemas—it usually works best to apply the majority of strokes in a

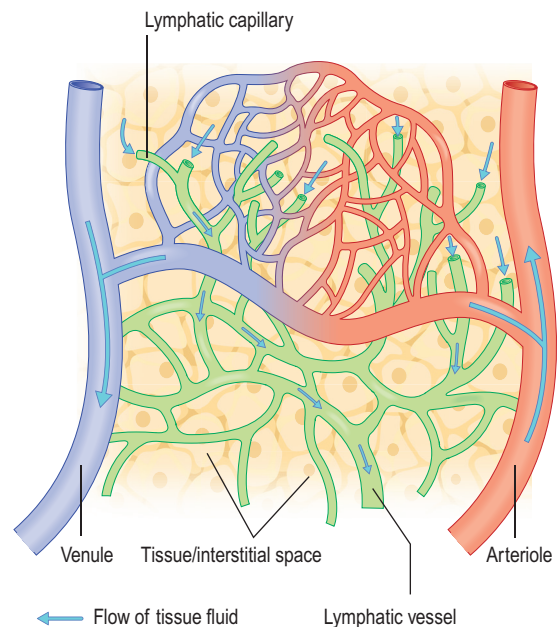


Figure 9.1

Diagram showing the flow of water from the ground substance to the ends of the lymph capillaries. Note that the majority of the ground substance water departs via the small venules, rather than the lymph pathway.

Illustration: U.S. National Cancer Institute's Surveillance, Epidemiology and End Results (SEER) Program.

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proximal direction towards the larger lymph vessels. This is because a large proportion of the inherent water is bound to inflammatory cells, cellular debris or pro-inflammatory cytokines, which then makes it difficult for those structures to enter the much smaller openings of the venules. In fact, several pathological tissue conditions, such as in systemic sclerosis and rheumatic diseases, have been associated with dysfunctional lymph flow dynamics. A beneficial therapeutic effect has been demonstrated for these conditions with lymphatic drainage applications (Schwartz et al., 2019).

An easy and free passage of water from the small arterioles into the ground substance, and from there back into the central bloodstream (via the small venules or lymph vessels) is an indication of healthy tissue condition (Figure 9.2). By contrast, an increasing inertia and stagnation of this fluid dynamic has been associated with a lower state of health, such as in metabolic syndrome or aging. Besides the previously mentioned lymphatic activation techniques, an enhancement of microcirculation has been recommended. This may occur via tool-assisted passive applications or via active exercise.

A novel aspect in this field are the “conduits” through the ground substance, as described by Weigelin et al. (2012). With the aid of third generation harmonic microscopy, which permits observation of cell migration *in vivo*, these researchers demonstrated that migrating cells tend to use the path of least resistance and thereby follow and widen specific pathways (conduits) through the ground substance. These conduits seem to be identical to the Primo Vascular System, previously called Bonghan channels, due to their size, distribution and high content of hyaluronan inside. While early discoverers of these channels associated them with acupuncture meridians,



Figure 9.2

Peripheral oedema is characterized by “pitting”, i.e. the persistence of indentation after a moderate pressure is applied to a small area and then released. This persistence expresses a restricted water renewal in the ground substance.

Photograph with the permission of James Heilman.

this topographical and functional association seems less clear today. However, the existence of these pathways as a third fluid transport system, next to blood and lymph vessels, is increasingly recognized (Soh et al., 2012). Increasing free and unobstructed movement through these conduits via active movement or therapeutic manipulation has been suggested to enhance immune system functioning (Swartz, 2018).

The miraculous capacities of hyaluronan

The ground substance is mainly composed of proteoglycans, most of which are hydrophilic (i.e. water loving), and of bound water. Similar to the structure of a bottle brush, or of plant

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mosses, the proteoglycans are arranged in a geometrical manner, thus offering the largest possible surface area for water molecules to attach to. Within the proteoglycan aggregates, hyaluronan often serves as a core protein, to which glycosaminoglycans are connected (Figure 9.3A). Hyaluronan is one of the most hydrophilic molecules in nature: it can trap up to 1000 times its own weight in water! Therefore, a high concentration of hyaluronan tends to increase the water content of a tissue.

One of the most exciting discoveries in the field of fascia science in recent years has been the reported existence of a new connective tissue cell type, which seems to be primarily focused on a rapid production of hyaluronan. A team of researchers at Padova University suggested the name “fasciocyte” for these cells, which they

describe as expressing a rather round cell shape, in contrast to the spindle-like shape of regular fibroblasts. They also showed that this cell type is frequently found in the upper and lower portions of loose connective tissue layers, i.e. at their transition to denser fascial layers adjacent to them (Stecco et al., 2018).

Hyaluronan is usually considered to be a lubricant, indicating that it decreases friction between adjacent tissue layers. This function is supported by a recent histological study, in which it was shown that the concentration of fascial tissues, which are exposed to a large degree of shearing/sliding motions, express hyaluronan concentrations of up to 10-fold higher compared with fascial tissues that are exposed to very little deformation (Figure 9.3B). This suggests that providing a fascial region with regular shearing

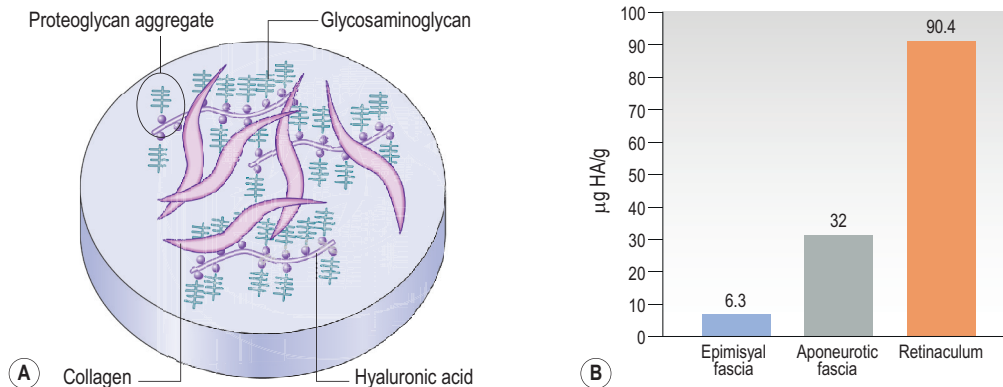


Figure 9.3A

Hyaluronan (also called hyaluronic acid) often serves as a core protein for the attachment of glycosaminoglycans within the ground substance.

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Figure 9.3B

Mean concentrations of hyaluronan (HA) in different fascial tissues in the human body. The concentration in epimysial fasciae (here around trapezius and deltoid muscles) is lower compared with aponeurotic fasciae (fascia lata and rectus abdominis sheet), while the retinaculum (here in the ankle) expresses by far the highest concentrations.

Illustration based on data from Fede et al., 2018.

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motions could induce a higher hyaluronan concentration in this region. To date, no studies have compared the value of regular stretches of the muscles and fasciae in the face with the reportedly beneficial effect of external cosmetic hyaluronan injections. However, given these general interactions between shearing motion and the production of hyaluronan within the tissue itself, it would not be surprising if the “internal treatment” could at least compare with the external one.

Interestingly, hyaluronan can also function as a “sticky glue”, preventing easy sliding between adjacent tissue layers. This apparently happens when hyaluronan takes on the form of super molecules, which are multiple times larger than in the usual molecular condition of this substance. There are indications that this glue-like condition, also described as an increase in viscosity, tends to happen more frequently when there is an acidic condition in the ground substance. It also happens when the tissue is exposed to repetitive mechanical overloading, such as in exercise or in repetitive strain injuries. While these changes may lead to a local stiffening and a decrease in range of motion in everyday living, an increase in tissue temperature tends to break down the large molecular structure into smaller fragments, which then express a much lower viscosity than in the previous condition (Pavan et al., 2014).

Similarly, it has been shown that the glue-like viscosity can be easily reduced with appropriate mechanical loading. While the application of sudden pressure tends to be “ignored” by the tissue, it has been shown that shearing motions, which induce a twisting/bending inside of the fibrous architecture, together with gradual redistribution of internal pressures, tend to induce a significant decrease of viscosity. This might

explain why immobility reduces fascial gliding and, consequently, range of motion. It may also explain the beneficial effects of many therapeutic myofascial release treatments (Pavan et al., 2014). A similar mechanism may also be at work when experiencing a beneficial effect in terms of a reduced tissue rigidity induced by regular movement practices in daily life. The beneficial response of such “warming up” exercises may then be partly comparable to the well-known response of shaking a ketchup bottle, which induces a decreased viscosity (or increased fluidity) in its content.

Soaking up the water

A healthy body will prevent the hyaluronan (and other hydrophilic elements within the ground substance) soaking up excessive amounts of water, since this would go along with a dramatic expansion of the total volume of the respective tissue. This healthy restraint is achieved by a constantly pre-stretched condition of the local collagen fiber network, which prevents the proteoglycan inducing an exaggerated tissue expansion (Figure 9.4). A simplified description is that in a healthy body condition, the proteoglycans are always “thirsty”. It is only in the case of injury, or other pathological changes, that they can soak up as much water as they would like to and then expand beyond their previous restrained condition. This can be easily observed in a fresh ankle sprain injury: here, the clearly visible tissue swelling often occurs during the first few minutes.

Based on this consideration, it appears rather unlikely that one could influence the water content (and pre-stretch) within dense fascial tissues just by drinking more water during the day. In other words, the water content in your ankle retinaculum will be primarily regulated by the

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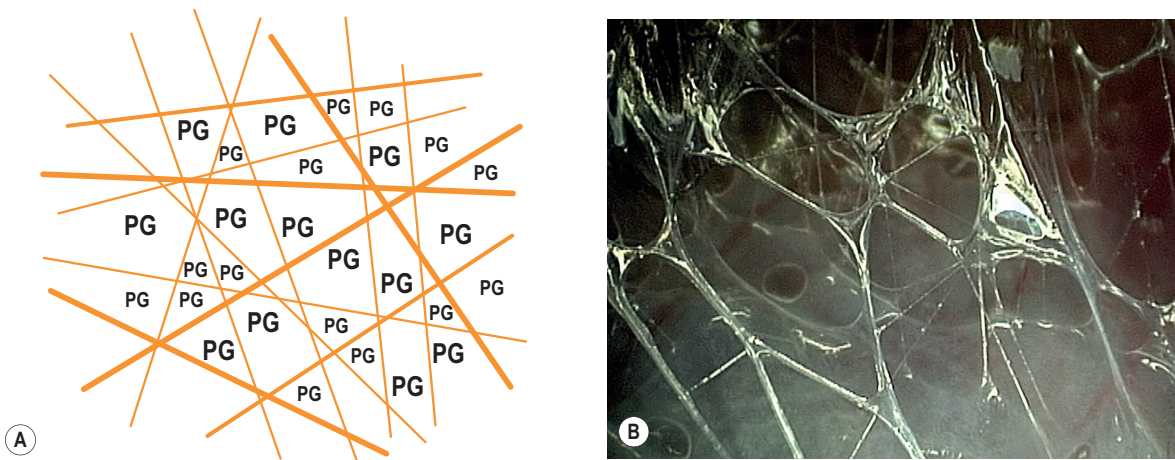


Figure 9.4

(A) Simplified model of the restraining action of collagen fibers (in pink) on the water uptake of the proteoglycans (PG) in the ground substance. In an uncontrolled condition, the proteoglycans would soak up a water volume several times their own size, which would result in tissue swelling. However, in healthy tissue, the potential swelling (and water uptake) is restrained by the constant pre-stretch of the fibrous network surrounding these “constantly thirsty” proteoglycans. (B) Photographic imaging of fresh fascia taken during endoscopic surgery reveals a similar pre-stretched relationship of the visible collagen fibers.

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pre-stretch of the local collagen network. Once that network expands, as in a fresh injury, the proteoglycans will soak up as much water from the arterioles as they want to, independently of how much water the person had been drinking that day. Another consequence of the described pre-stretched situation in healthy tissues is that any additional uptake of water into the ground substance will probably happen via an altered fluid-pull (suction) from within the ground substance, not by an increasing supply (push) of water from the outside into the pre-stretched tissue.

Squeezing a sponge: Dynamics of dehydration and rehydration

When exposed to mechanical loading, the water content in a fascial tissue tends to be decreased: similar to the deformation of a wet sponge, once it is either stretched or

compressed, some of the internal water content will be squeezed out during the loading condition. Similarly, after the loading is stopped, the sponge is expected to rehydrate again, with either the same water or with new water from within the vicinity. Our laboratory at Ulm University demonstrated this sponge-like dehydration and rehydration effect multiple times in an organ bath condition (Schleip et al., 2011). We also showed that the loading-induced dehydration tends to go along with a temporary loss of tissue stiffness (at least in ligamentous tissues), and that the subsequent rehydration tends to restore the previous tissue stiffness again (Figure 9.5). A similar effect has been documented with a foam roller-like myofascial treatment on the plantar fascia (Frenzel et al., 2015) and with therapeutic application of a Rolting myofascial release technique on the lumbar fascia (Dennenmoser et al., 2016).

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Interestingly, our organ bath experiments indicated that a “supercompensation” effect can be achieved if the magnitude of the tissue loading is large and the subsequent rest period is sufficiently long. This “strain hardening” could be observed in some cases, when a subsequent increase of stiffness beyond the original condition was induced. If the supercompensation could also be shown *in vivo*, it might provide future applications for preconditioning routines in athletic performance conditions (Schleip et al., 2011). Because hydration changes can impact the failure stress of at least some fascial tissues (Werbner et al., 2019), a more detailed investigation of the various interactions between different mechanical loading protocols, resultant hydration changes and subsequent effects of biomechanical tissue properties provides a very promising area in current research.

Liquid crystals within us

Ubiquitous on earth, water is present in all life forms. However, recent research has revealed that the water inside of living bodies exposes very surprising properties, which are certainly different to what has so far been known about regular bulk water.

The basic idea that “interfacial water layers”—i.e. the arrangement of water molecules in the vicinity of biological surfaces—play a fundamental role in biological systems was proposed in a visionary paper by Szent-Györgyi (1971). As Gruebele, one of the leading scientists in this area, explains: “Water in our bodies has different physical properties from ordinary bulk water, because of the presence of proteins and other biomolecules. Proteins change the properties of water to perform particular tasks in different parts of our cells. Water can be viewed as a ‘designer fluid’ in living cells” (University of Illinois at

Urbana-Champaign, 2008). Apparently, the interaction of water molecules with hydrophilic and hydrophobic biomolecules in their vicinity influences their behavior in very surprising manners. Gruebele states: “We previously thought proteins would affect only those water molecules directly stuck to them ... Now we know proteins will affect a volume of water comparable to their own. That’s pretty amazing” (University of Illinois at Urbana-Champaign, 2008).

Pollack (2013) examined the behavior of what he calls “vicinal water” within the temporomandibular joint and described it as a crystalline architecture. While the water molecules in this state still vibrate very rapidly in this condition, they do so within very stable conditions, which are called “liquid crystal” in physics (Pollack, 2013). Because of the previously described “bottle brush architecture” of the proteoglycans in the ground substance of fascial tissues, a large proportion of the water molecules apparently take on this special crystalized condition. As Pollack (2013) states, “The combined data from three different methods lead to the conclusion that all or almost all of the water in the intact disc is bound water and does not have properties consistent with free or bulk water.”

The crystalized water exposes very different properties to regular bulk water in terms of a significantly different density, an increased viscosity, different light transmission and a different electrical conductance (Sommer et al., 2011). Note that while most of the proteoglycans are hydrophilic, the small elastin fibers are hydrophobic. Their strong water repulsion then induces a coating-like accumulation of specially arranged water molecules around them, in which these molecules’ binding sites face away from the fibers, and thereby take on a crystalized condition not unlike the vicinal water around hydrophilic

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surfaces. While it is a common assumption that morphological differences in aging skin are due to a change in the elastin fibers, Sommer et al. (2008) demonstrated that the amount of water coating around these fibers plays a major role. While the elastin in young fibers is surrounded (and buffered) with a very thick zone of crystalized water, this coating tends to get thinner and thinner as we age, due to the accumulation of free radicals and other metabolic waste products in the small vacuum-like zone between the fibers and their coatings. In an intriguing experiment, Sommer and Zhu (2008) showed that after an attempt to push the “dirty bulk water” away

from this zone using a special laser, the elastin fibers were apparently surrounded again with thicker coatings of crystalized water and the skin in these regions took on an obviously more juvenile appearance (Figure 9.5).

In a personal discussion with our department, Dr Andrej Sommer and his Rolfer colleague, Kai Hodeck, expressed an assumption that a mechanical sponge-like myofascial treatment, in which the inherent water is steadily pushed into different directions, could potentially exert a similar (or even stronger) renewal effect on these water coatings than the one reported in

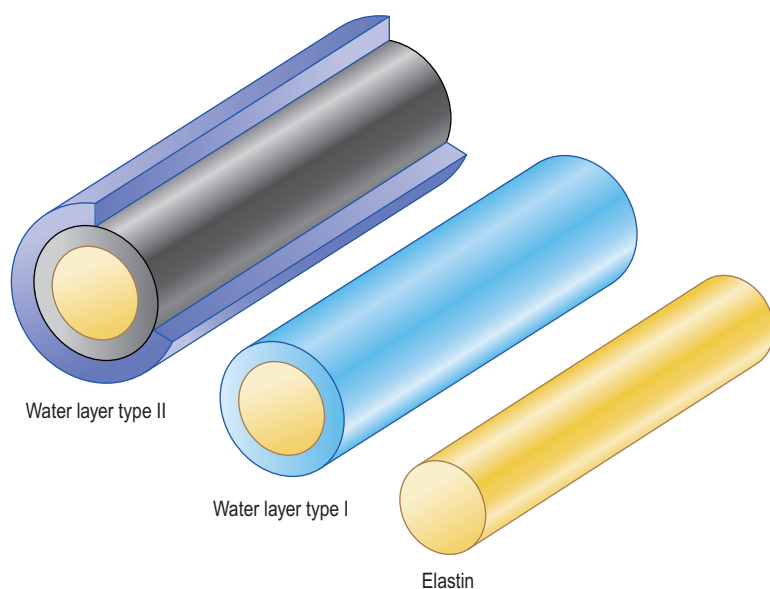


Figure 9.5

Elastin fibers are hydrophobic, i.e. water repellent (right). In their native condition, the elastic fibers are surrounded by a thick layer of crystalized water layer around it (middle), which protects the structural integrity and stretch-ability of the fibers. During the process of physiological aging (left), elastin fragmentation and disintegration are probably induced by metabolic waste products (black), which inhibit the formation of the crystalized water coating. The process of aging can therefore be understood as a decrease of the protective water coating. Therapeutic intervention—via a special laser application—was apparently able to reverse some of that aging process, i.e. to shift the water environment from the situation shown on the left more towards the “younger condition” shown in the middle.

Illustration modified after Sommer and Zhu (2008).

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their study. If this speculation were supported and validated by future investigations, it would mean that regular stretching, foam rolling or similar treatments could induce a higher proportion of crystalized water in the ground substance and, thereby, induce and exert an anti-aging effect on the tissues. It will be exciting to follow the ongoing research of these pioneers and their colleagues in the next few years.

Clinical summary

Water constitutes the majority of the volume of our fascia. Any change in this element can be

expected to exert significant effects on the whole tissue. For movement therapists, it is helpful to think of the sponge-like changes of their interventions. A skilful interplay of temporary dehydration and subsequent rehydration promises not only a renewal of the tissue, but may also change the stiffness of the treated area. Applying mechanical pressure may also induce a change of the viscosity via a molecular change of hyaluronan. Lack of motion, on the other side, will lead towards a more rapid aging effect, which probably is associated with a decrease of crystalized water within the tissue.

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